# Omar Elgeoushy

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## EDUCATION

**Master of Science in Electrical and Computer Engineering** August 2023 - May 2025
**Purdue University,** Indianapolis, IN

**Bachelor of Science in Computer Science** Graduated December 2022
**Virginia Tech**, Blacksburg, VA
GPA: 3.48/4.00 | Mathematics Minor | Dean's Honor List: Spring 2021, Fall 2021 and Spring 2022

## SKILLS and Certificates

**Programming Skills**: Java, Python, C, C++, C#, Ruby, React, JavaScript, Angular, MATLAB, CSS, Spring, SQL, PyTorch, .NET, Swift
**Networking Skills:** Broadband Network, TCP/IP, PON, SDN, VPN, VLANs, SNMP
**Other Skill:** Github, Jira, GDB, Valgrind, Linux, Unix, AWS, Data-structure, Scrum, AutoGPT, Agile, Waterfall, AI, Docker
**Language Skills:** Fluent in English, German, and Arabic
**Certificates:** AWS Cloud Practitioner (In Progress)

## WORK EXPERIENCE

**Software Engineering Research Assistant at Purdue**, Indianapolis, IN August 2023 – Present
Autonomous Off-Road Ground Vehicle Simulator:

- Developing a Python-based AI model by utilizing a dataset of over 2500 images with TensorFlow and DeepLab to facilitate object detection and semantic segmentation within the autonomy stack, a pivotal element for self-driving cars
- Integrating the model seamlessly into the Multibody Dynamics & Terramechanics (IVRESS) system, thereby advancing autonomous vehicle technology
- Utilizing Unreal Engine 5 to visualize testing, enhancing the practical application of the technology

**Software Engineer Intern at Si-Ware Systems**, Menlo Park, CA June 2023 – September 2023

- Developed a Python-based customer support chatbot using the AutoGPT API, streamlining support processes
- Developed automated assistance and enhanced communication to deliver a seamless user experience on the front-end with React integration based on custom data fed to the model
- Utilized Pinecone to implement a vector database, improving search and retrieval processes for the chatbot

**Software Engineer Intern at 5APlus NetArch**, Cheyenne, WY Summer 2021 and 2022
Network Monitoring Dashboard: June 2022 – August 2022

- Created a Python-based network monitoring dashboard using SNMP and TCP/IP to analyze data from network devices
- Implemented Angular and CSS for a user-friendly network performance monitoring interface
- Stored network data in a MySQL database for easy retrieval and analysis

Finance App: June 2021 – August 2021

- Contributed to the development of the Finance app, enabling payment processing using cryptocurrency for users
- Collaborated with the development team to implement and test features for the app
- Focused on backend development using Springboot and utilized SQL to manage and manipulate the database

**Undergrad Teacher Assistant (UTA) | Computer Science Department at Virginia Tech** August 2022 – December 2022

- Assisted 300+ students in achieving success in CS course and providing tutoring to students who needed extra support

## PROJECTS

**Case Study Library:** August 2022 – December 2022

- Designed and developed a website hosted on AWS to serve as a resource for CS students at Virginia Tech, utilizing various AWS tools including S3 and DynamoDB
- Implemented an upload page that allows students to upload their own presentations to the website
- Senior project website featuring over 850 presentations for viewing, accessible at https://casestudies.cs.vt.edu/